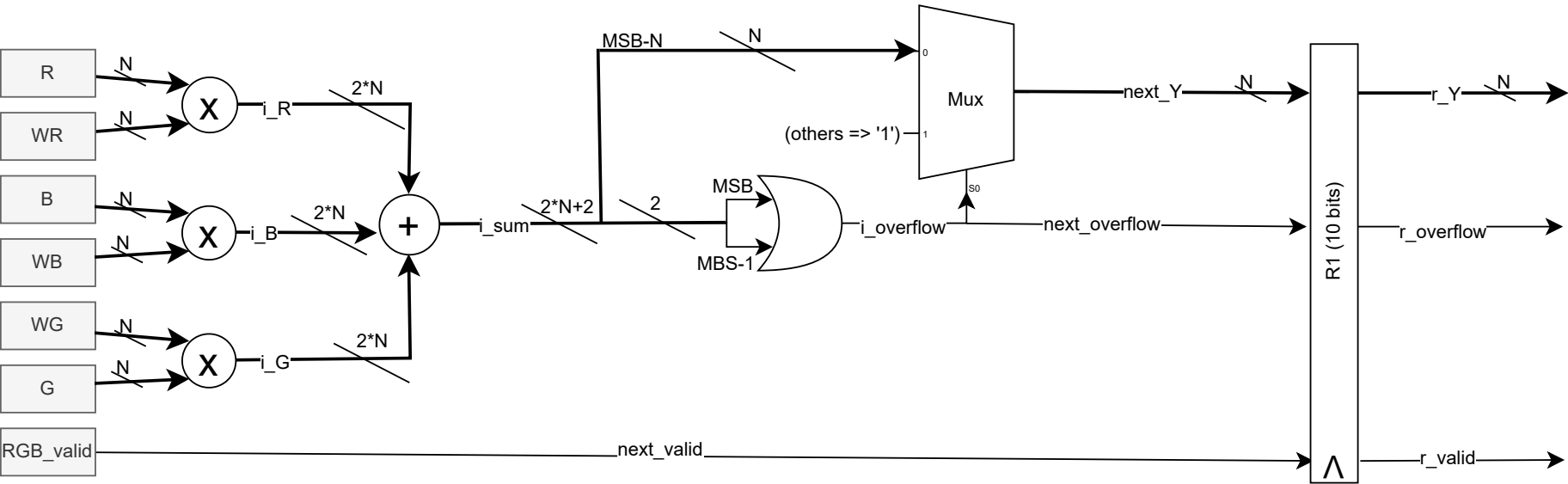


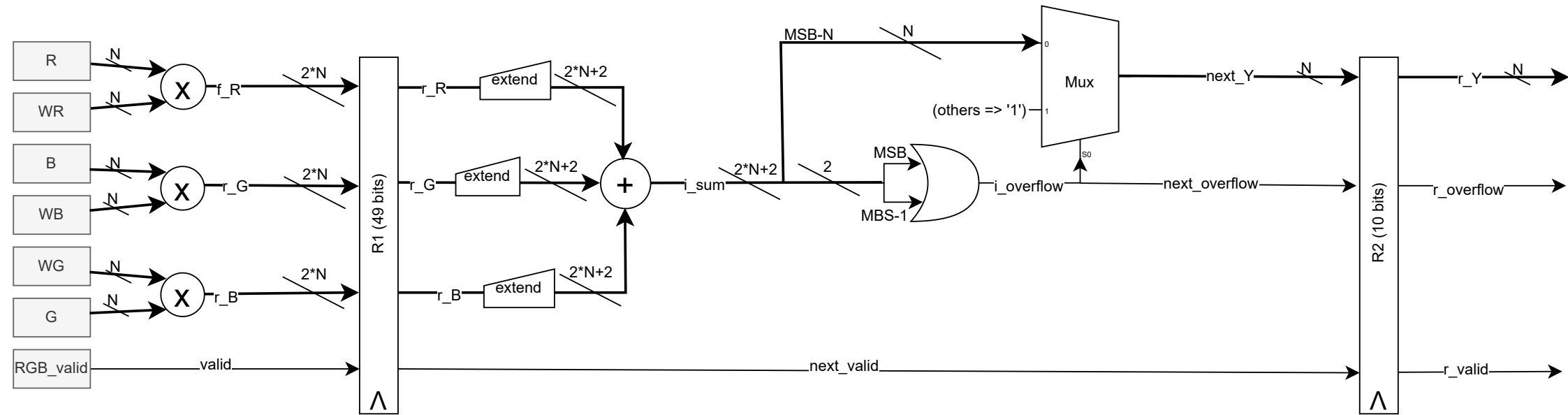
Datapath for existing grayscale

(N+1+1=10 flipflops)



Datapath for two-stage pipeline

(49+10=59 flipflops)



Step 1

Step 2

TASK 1

- Q1: the sum of 8 bits a and 8 bits b will give an answer of 9 bits
- Q2: the product of two 8 bits numbers a and b will give an answer of 16 bits
- Q3: adding a,b and c with 8 bits will give a result with 9 bits
- Q4: the sum of four 8 bits numbers gives a result of 10 bits
- multiplying the sum with an 8 bits number e gives a result of 18 bits